

GROUP 16

PLATO

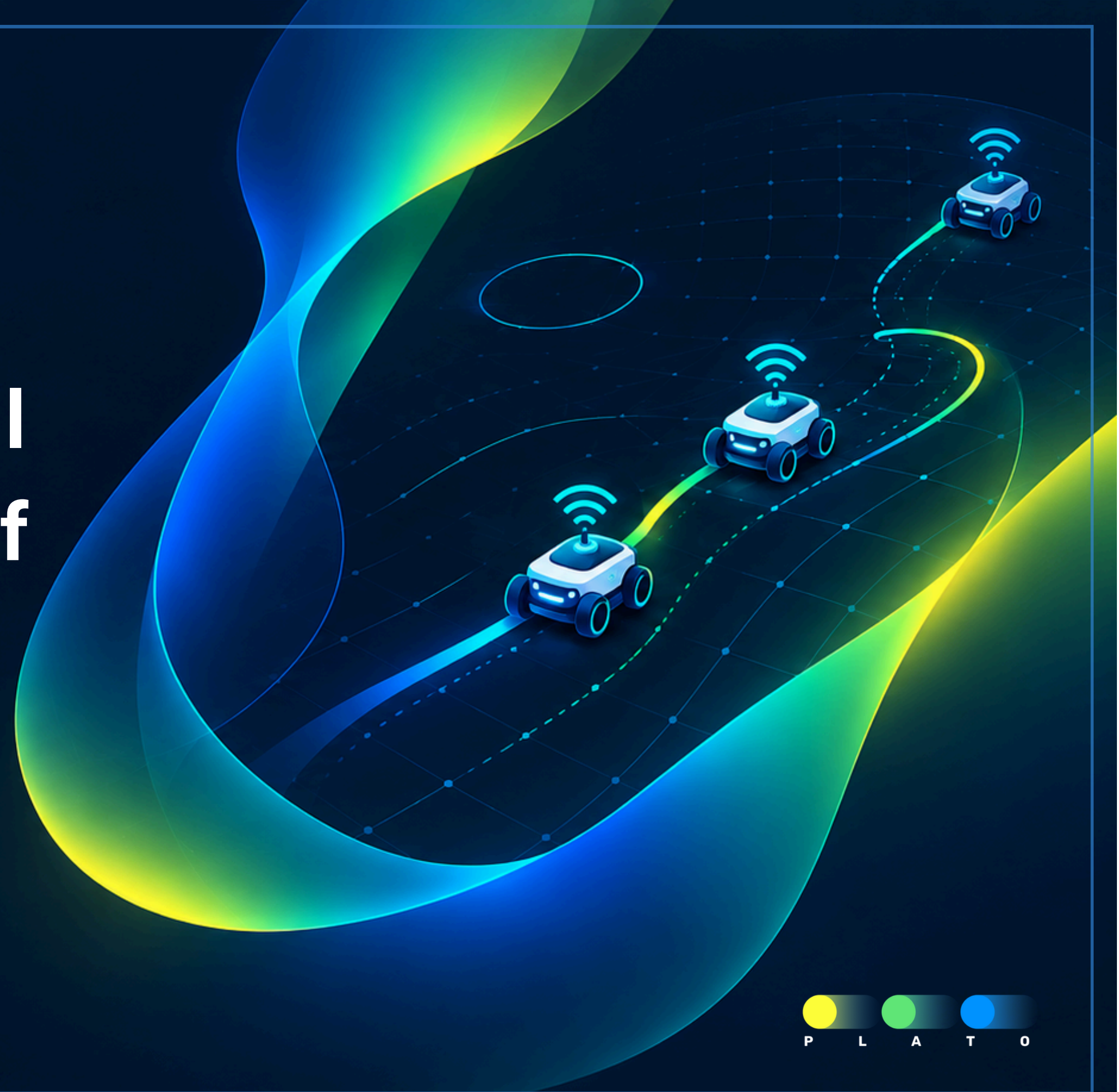
Autonomous Control and Teleoperation of Robot Platoons

In Pursuit of the Ideal Path

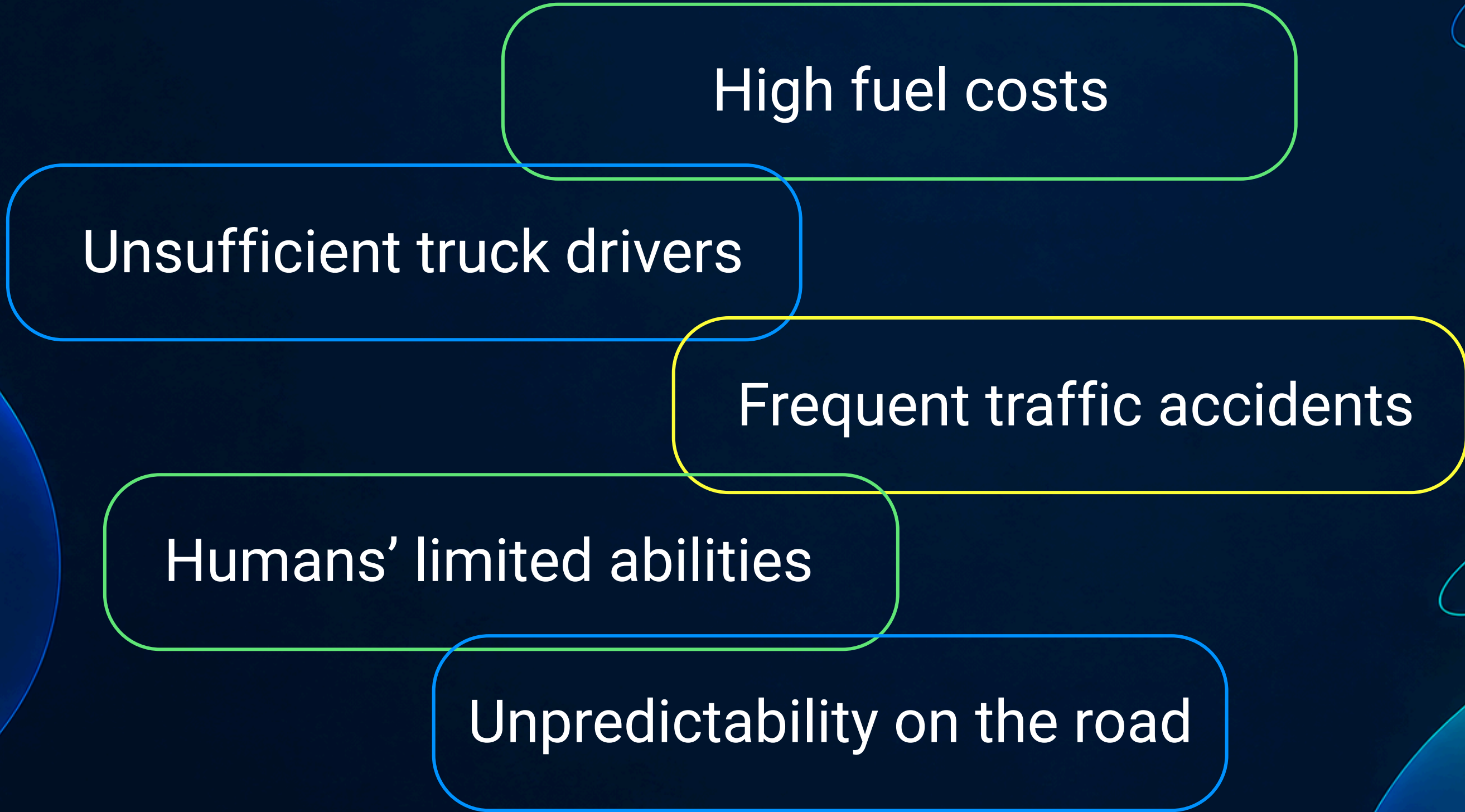


deti

universidade de aveiro
departamento de eletrónica,
telecomunicações e informática



Context



```
graph LR; A[High fuel costs] --- B[Unsufficient truck drivers]; B --- C[Frequent traffic accidents]; C --- D[Humans' limited abilities]; D --- E[Unpredictability on the road];
```

High fuel costs

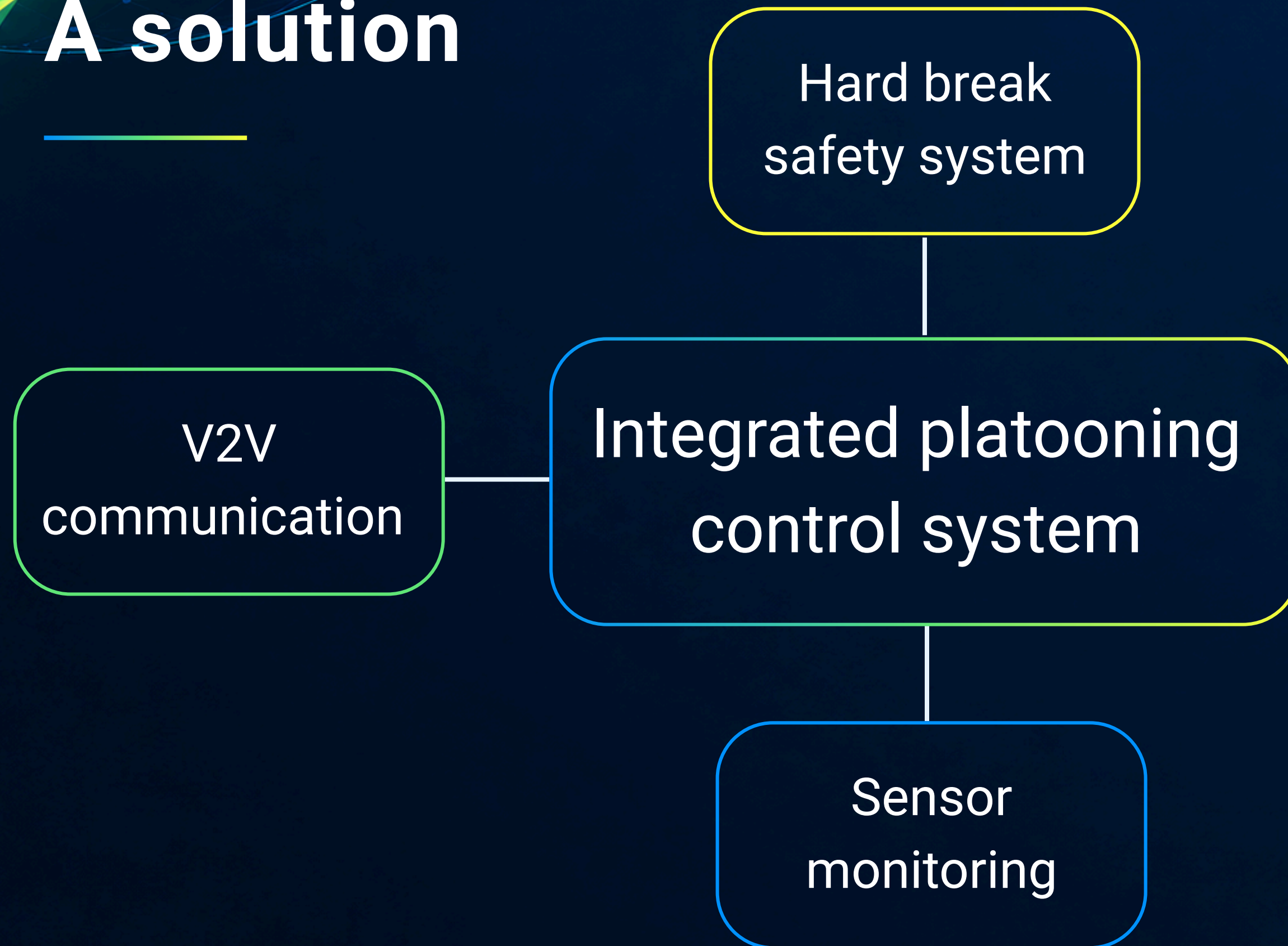
Unsufficient truck drivers

Frequent traffic accidents

Humans' limited abilities

Unpredictability on the road

A solution



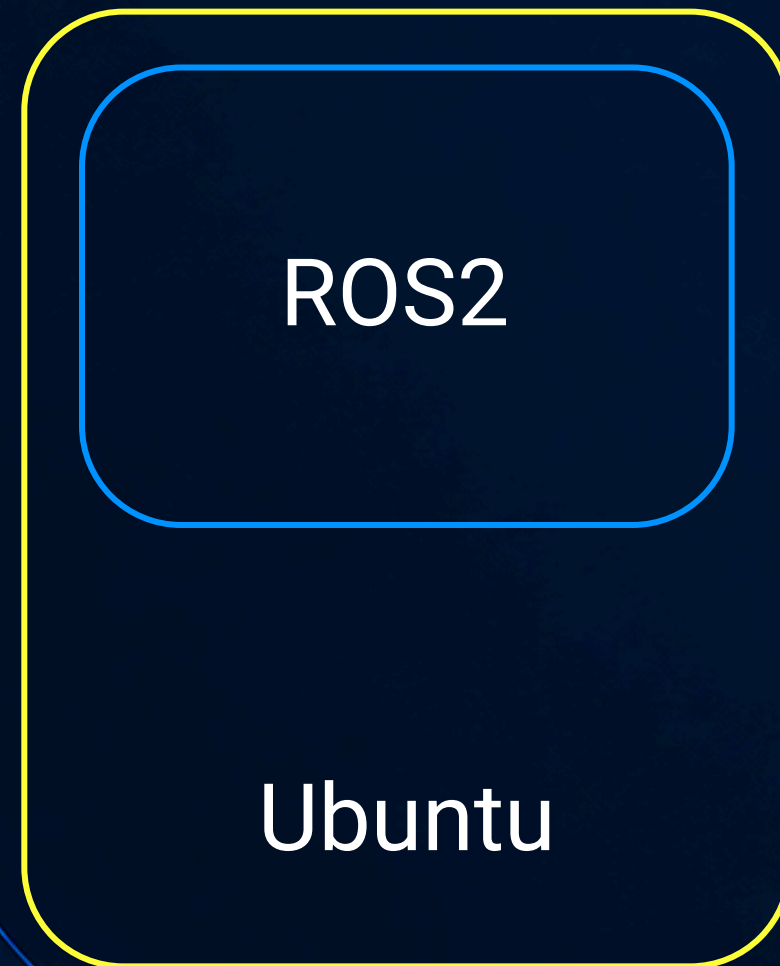
Our goals

Better efficiency and costs

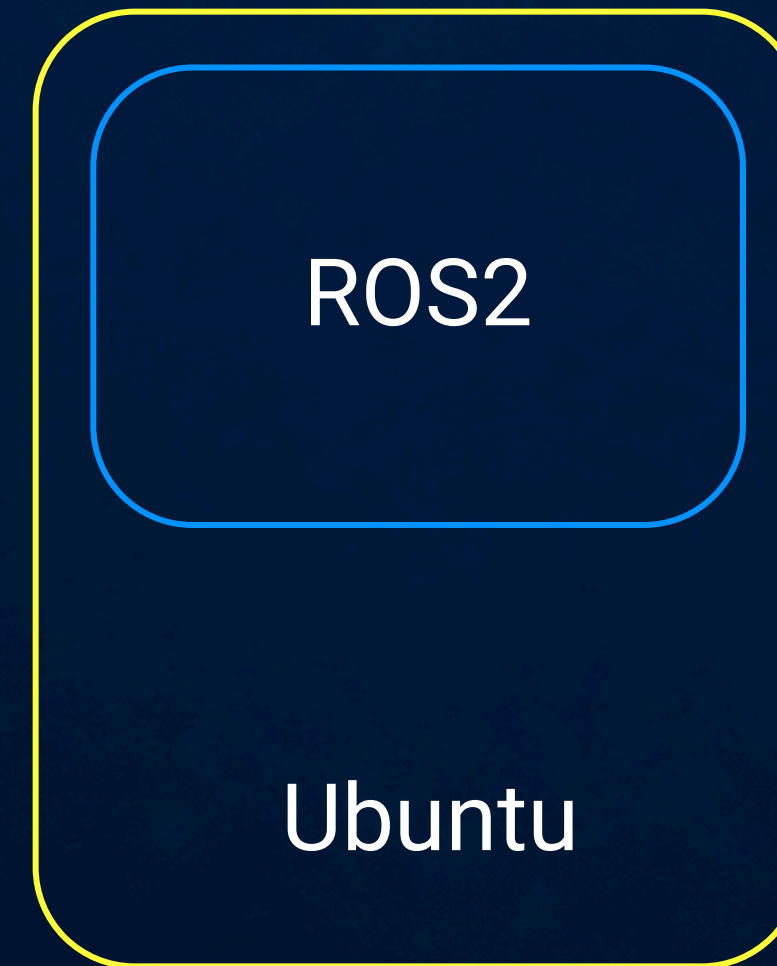
Improvement in road safety

Reduction of traffic accidents

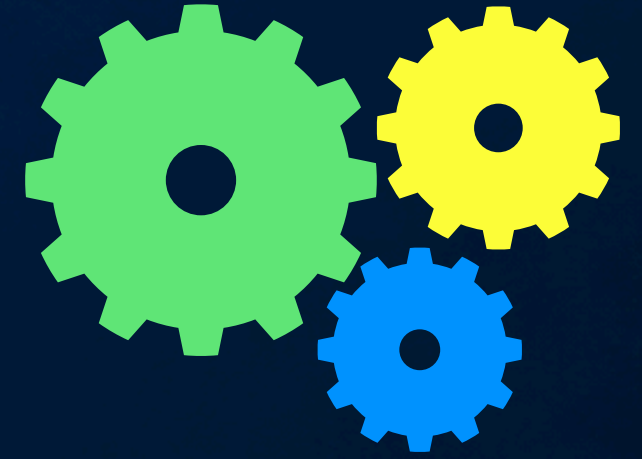
Architecture overview



Wi-Fi



Key features



Distributed Communication Architecture in ROS 2

Autonomous Navigation and Leader Teleoperation

Trajectory and Velocity Replication

Safety Scenario 1 – Hard Braking

Safety Scenario 2 – Platoon Split

Tasks

- | | | | |
|---|----------------------------|---|-----------------------------|
| 0 | Gathering the requirements | 5 | Leader following algorithms |
| 1 | Analysing the platforms | 6 | Hard break |
| 2 | Modeling and simulation | 7 | Platoon split |
| 3 | V2V model development | 8 | Testing and implementation |
| 4 | Teleoperations interface | 9 | Documentation |

The team

Henrique Reveles

João Neves

Matilde Sarabando

Pedro Rocha

Ricardo Rebelo

Main advisor

Joaquim
Ferreira

Co-advisor

João Pereira
Almeida

Contributor

Mohannad
Jooriah

Project Calendar

Week 1

15/09/2026

Study the ROS2 platform
Learn the basics
Solve some tutorials and courses
Understanding the project

Week 2

22/09/2026

Requirement gathering
Project calendar elaboration
Development of the microsite
Prepare the presentation for M1
Deepen the knowledge about ROS2

Week 3 - M1

29/09/2026

Architecture design
Revision about platooning
Communication V2V
Analysis of the platform AgileX
LIMO PRO

Week 4

06/10/2026

Creation of the multi-robot virtual environment in Gazebo
Cinematic model configuration
Prepare the presentation for M2

Week 5 - M2

13/10/2026

Business plan
Legal requirements
Technical risks
LIMO PRO sensors
Initial motion command validation

Week 6 - M2

20/10/2026

Developing of the ROS topics / messages for V2V short-range communication and implementation of the remote teleoperation station for leader's driver

Project Calendar

Week 7

27/10/2026

Developing of the ROS topics /
messages for V2V short-range
communication
Prepare presentation for M3

Week 8 - M3

03/11/2026

Developing of the ROS topics /
messages for V2V short-range
communication
Finish any work left from previous
weeks

Week 9 - M3

10/11/2026

Finish all work from previous weeks
Programming of the lateral
and longitudinal tracking algorithms
in follower robots using V2V and
LiDAR fusion

Week 10

17/09/2026

Programming of the lateral
and longitudinal tracking algorithms
in follower robots using V2V and
LiDAR fusion

Week 11

24/09/2026

Programming of the lateral
and longitudinal tracking algorithms
in follower robots using V2V and
LiDAR fusion
Document everything

Week 12

01/09/2026

Programming of the lateral
and longitudinal tracking algorithms
in follower robots using V2V and
LiDAR fusion
Finish MVP

Project Calendar

Week 13

08/12/2026

Scalability evaluation
Security analysis
Performance check
Machine Learning model validation
Prepare for presentation M4

Week 14 - M4

15/12/2026

16/12/2026

Finish any previous work
Get feedback from presentation to fix and improve whatever is needed
MVP testing and evaluation

Jan

01/2026

MVP review and improvement
Implementation of hard break mechanism when the leader breaks
Make the poster for Students@DETI

Feb

02/2026

Implementation of hard break mechanism when the leader breaks
Implementation of division protocol with transition to teleoperation

Mar

03/2027

Implementation of division protocol with transition to teleoperation
Integration testing with the real robots
Control and network parameter optimization

Apr

04/2027

Integration testing with the real robots
Control and network parameter optimization
Create the promotional video

May

05/2027

Final report
Final project presentation
Students@DETI

Related work

	Kratos	ENSEMBLE	CohdaWireless	Fernride	PLATO
Platooning	Yes	Yes	No	Yes	Yes
Lider	Human	Human	Autonomous	Autonomous	Autonomous
Followers	Human safely rider	Human	Autonomous	Autonomous	Autonomous
Tele-operation	No	No	Yes	No	Yes
hard brake/split	No	Yes	No	Yes	Yes
Scale	Real trucks	Real trucks	terminal trucks	Cars on a track	small vehicles(LIMO)

Thank you!



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